hedged_blend

portfolio strategy | family: portfolio | status: validated | derived from `s9b_inverse_vol`

What this strategy does

Market-neutral inverse-vol equity book (0.8x) + corrected-quote Kelly options book (1.6x), on one NAV

Why it should work. The sleeves are uncorrelated with each other (about -0.05) and with the market (~0 / 0.18): equity alpha earned most of its return in 1996-2003 and made +10% in 2008 when the options book lost; options earned most after 2003. Weights set so each sleeve contributes equal stand-alone risk, scaled to ~5% portfolio volatility.

Design notes (from the strategy's source file).

PortfolioStrategy: a fixed-weight, daily-rebalanced blend of other registered strategies.

A sleeve weight is a multiple of fund NAV allocated to that strategy's own sizing: weight 1.0 runs the strategy exactly as backtested on the whole NAV, 0.8 runs it 20% smaller. Weights need not sum to one -- the sleeves are low-correlation books that each use only part of their capital (the options sleeve spends at most its premium budget, the equity sleeve is margin-financed long/short), so they can share one NAV. Returns compound daily: r_p = sum(w_i r_i), NAV_t = NAV_{t-1} (1 + r_p).

Sleeve results come from the strategy registry itself (re-run in-process, ~seconds), so the blend always reflects the current sleeve definitions. Days with no sleeve activity (options in 2011, where the source file is corrupt) contribute zero for that sleeve.

parameter	value
w_s9b_inverse_vol	0.8
w_opt_kelly_cap20	1.6

Performance

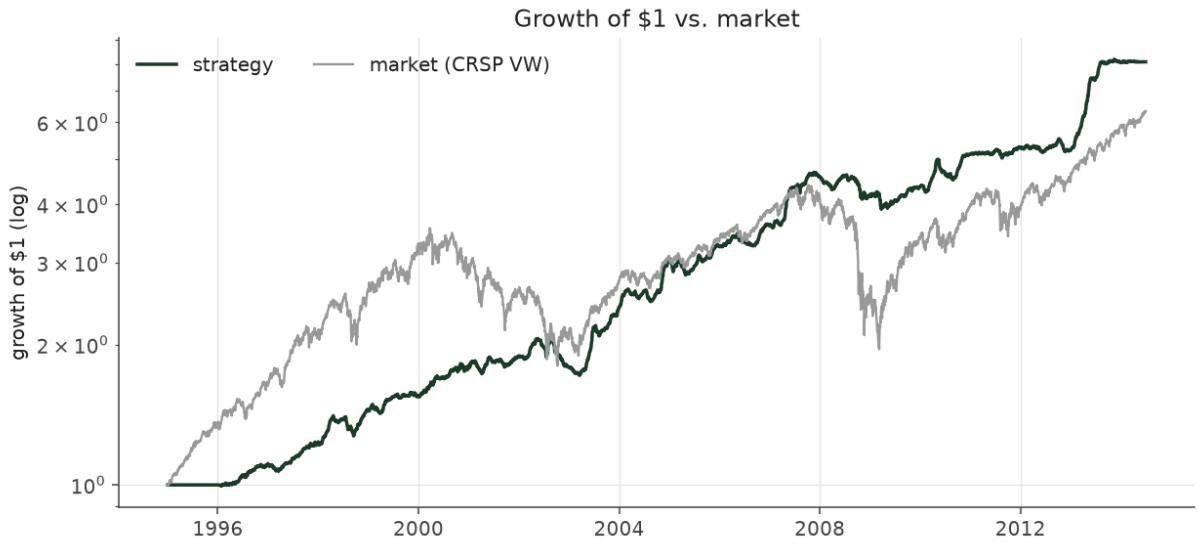
metric	hedged blend	market (CRSP VW)
Annualized return	11.33%	9.94%
Annualized volatility	4.83%	19.35%
Sharpe (NAV-pct)	2.25	0.59
Sortino	3.58	
Max drawdown	-16.7%	-55.5%
Calmar	0.68	
Final NAV ($10M start)	$80,879,071	
Worst day	-2.65%	
% positive days	54.0%	
Sharpe 1996-2001	2.41	
Sharpe 2002-2007	2.70	
Sharpe 2008-2013	1.91	

Correlation of daily returns with the market: **0.13**.

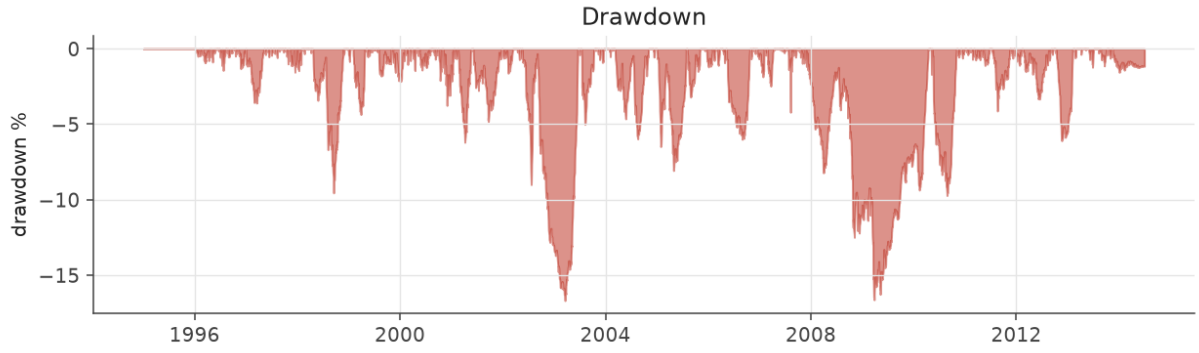
• best Sharpe among portfolio: `hedged_blend_s10` (2.5); this one: 2.25

• best return among portfolio: `hedged_blend_s10` (0.124); this one: 0.113

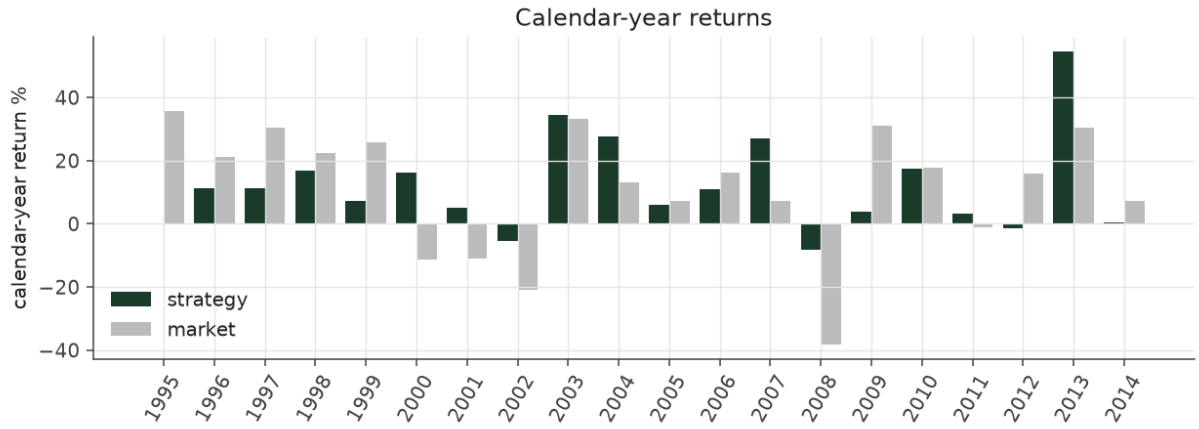
• best Calmar among portfolio: `hedged_blend_s10` (0.802); this one: 0.679



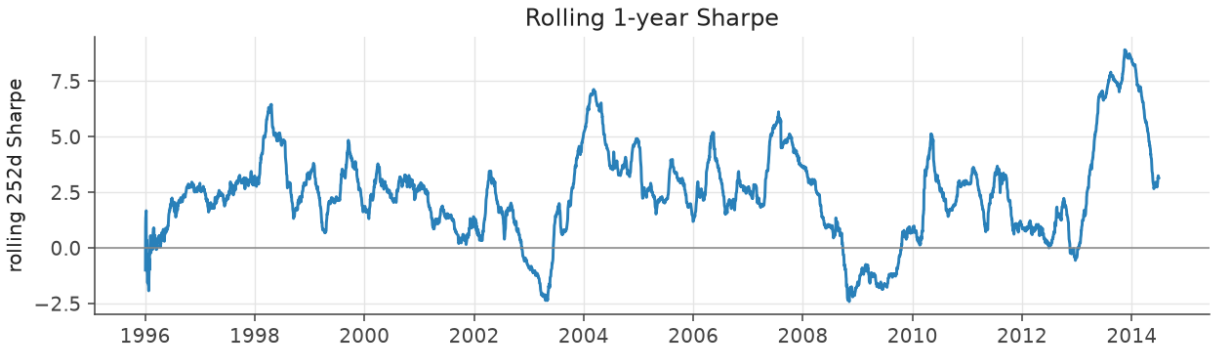
NAV (log scale)



Drawdown from running peak

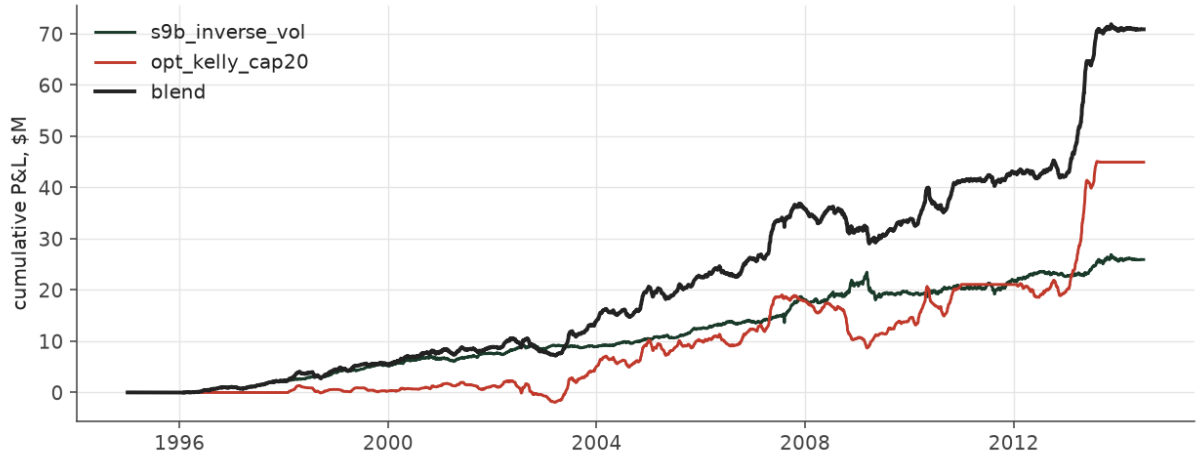


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Each sleeve's contribution to the blend's cumulative P&L

sleeve	weight	pnl M	share of pnl
s9b_inverse_vol	0.80	25.93	36.6%
opt_kelly_cap20	1.60	44.95	63.4%

• Correlation between the sleeves' daily returns: **-0.05**.

• Blend market beta 0.033, correlation to the market 0.13.

Return, volatility and Sharpe of the blend are in the Performance table above; the cross-strategy report (`python -m pead compare`) shows the full weight frontier, the walk-forward (out-of-sample) weights and the volatility-target table.